

Question:

A 17-year-old teenager is brought to the emergency department with severe bleeding from his right hand. He was involved in a gang fight about 30 minutes ago where he received a penetrating stab wound by a sharp knife in the region of the ‘anatomical snuffbox’. A vascular surgeon is called in for consultation. Damage to which artery is most likely responsible for his excessive bleeding?

Answer Choices:

- A. Radial artery
- B. Ulnar artery
- C. Princeps pollicis artery
- D. Brachial artery

Reasoning + Answer:

<think>
To determine which artery is most likely responsible for the excessive bleeding , we need to consider the location of the injury . The " anatomical snuffbox " is a triangular region on the radial , dorsal aspect of the hand , extending from the radial styloid process to the scaphoid bone . This region is innervated by the superficial branch of the radial nerve and has a significant concentration of blood vessels , including the radial artery . The radial artery runs through the anatomical snuffbox and is closely related to the scaphoid bone , making it a prime location for trauma-related bleeding . Considering the nature of the injury and the anatomy of the hand , the most likely artery involved in the bleeding is the radial artery , which is a major artery that runs through the anatomical snuffbox and is prone to injury in such cases.</think>
<answer> A. Radial artery

